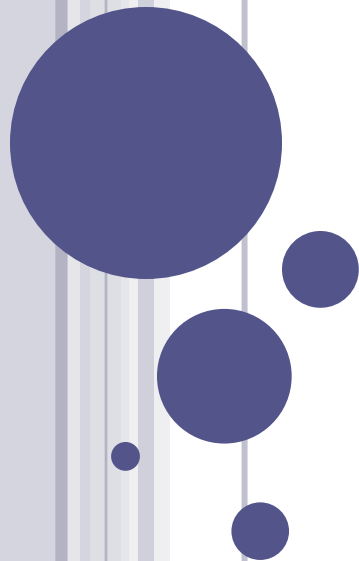


SOFTWARE ENGINEERING

SUBJECT CODE: MCAXX11509

UNIT 1 : INTRODUCTION TO SOFTWARE ENGINEERING



UNIT OUTLINE

- Introduction to Software Engineering
- Software Engineering Approach
- Software Process and it's characteristics.
- Software Development Life Cycle .[SDLC]
- Software Development Process Model .
 - A. Waterfall model
 - B. Prototyping model .
 - C. Iterative Enhancement model(Incremental model)
 - D. Spiral model .



Introduction Of Software Engineering

- The term is made of two words, software and engineering. **Software is more than just a program code.** A program is an executable **code, which serves some computational purpose.**
- Software,** when made for a specific requirement is called **software product.**
- **Engineering on the other hand,** is **all about developing products, using well-defined, scientific principles and methods.**
- Software engineering is an engineering branch associated with development of software product using well-defined scientific principles, methods and procedures.**



- The **outcome** of software engineering is an **efficient and reliable software product**.
- Software engineering is the **process of designing, building, and maintaining software applications** in a structured way.
- It involves several key steps:
 - first, **understanding what users need from the software**;
 - then **planning** how the software will work;
 - next, **writing the code** to create the software;
 - followed by **testing** it to make sure it **works correctly**;
 - and finally, keeping it updated and fixing any issues that arise.



- This discipline uses specific methods and tools **to help teams work together efficiently and manage their projects well.**
- The **goal of** software engineering is **to produce high-quality software that is reliable, easy to use, and can grow or change as needed over time.**



SOFTWARE ENGINEERING APPROACH

1. **Organized Process:** Use clear and structured methods like Agile, Waterfall, to make work predictable and consistent.
2. **Grow and Update Easily:** Build software that can grow with the user's needs and is easy to fix or improve later.
3. **Focus on Quality:** Check and test the software carefully at every step to ensure it works well.
4. **Smaller Parts:** Divide the software into small, simple pieces to make building, testing, and fixing easier.
5. **User-Friendly Design:** Make sure the software is easy to use and solves the user's problem.



6. **Plan for Risks:** Find potential problems early and have plans to handle them, like limited resources or technical issues.
7. **Keep Good Records:** Write down everything clearly during the project, like plans, designs, and how things work, so it's easy to understand later.



SOFTWARE PROCESS AND IT'S CHARACTERISTICS

1. What is a Software Process?

- A software process is a **set of activities and related tasks** that are carried out to develop and maintain software systems.
- It defines the "how" of software development, providing a framework to produce high-quality software efficiently.

➤ Characteristics of a Good Software Process

1. Predictability:

- The process should give similar outcomes every time, making it **easier to plan time and resources**.



2. Repeatability:

- Following the same steps should give the same results if the conditions are the same..

3. Efficiency:

- should save time, money, and effort while avoiding waste.

4. Quality-Oriented:

- Focuses on producing software that meets customer requirements

5. Scalability:

- Can handle projects of varying sizes and complexities, from small applications to large systems.

6. Flexibility:

- Adapts to changes in requirements, technology, and market conditions without major disruptions.



7. Transparency:

- Everyone involved should be able to see how the project is going, including any challenges.

8. Risk Management:

- it should identify and handle problems early to avoid bigger issues later.

9. Documentation:

- Keep clear and complete notes at every step, so future updates and fixes are easier.

10. Incremental and Iterative:

- Build the software in small pieces, so feedback can be used to make improvements as you go.



Software Development Life Cycle .[SDLC]

- A software life cycle model (also termed **process model**) is a **pictorial and diagrammatic representation of the software life cycle.**
- A life cycle model **represents all the methods required to make a software product** transit through its life cycle stages.
- It also **captures the structure in which these methods** are to be undertaken.
- Different life cycle **models may plan the necessary development activities to phases in different ways.**

- Thus, no element which life cycle model is followed, the **essential activities are contained in all life cycle models though the action may be carried out in distinct orders in different life cycle models.**
- During any life cycle stage, **more than one activity may also be carried out.**





➤ **Stage1: Planning and requirement analysis**

- Requirement Analysis is the **most important and necessary stage in SDLC.**
- The **senior members of the team perform it with inputs** from all the **stakeholders and domain experts** in the industry.
- **Planning for the quality assurance requirements and identifications of the risks associated with the projects** is also done at this stage.
- **Business analyst and Project organizer** set up a meeting with the client to gather all the data like what the **customer wants to build, who will be the end user, what is the objective of the product.** Before creating a product, **a core understanding or knowledge of the product is very necessary.**



➤ **Stage2: Defining Requirements**

- Once the requirement analysis is done, the **next stage** is to **certainly represent and document the software requirements** and get them accepted from the **project stakeholders**.
 - This is accomplished through "SRS"- Software Requirement Specification document which contains **all the product requirements to be constructed and developed during the project life cycle**.
- ## ➤ **Stage3: Designing the Software**
- The next phase is about to bring down all the knowledge of **requirements, analysis, and design of the software project**.
 - This phase is the product of the **last two, like inputs from the customer and requirement gathering**.



➤ **Stage4: Developing the project**

- In this phase of SDLC, the actual development begins, and the programming is built.
- The implementation of design begins concerning **writing code**. **Developers have to follow the coding guidelines described by their management and programming tools like compilers, interpreters, debuggers, etc.** are used to develop and implement the code.

➤ **Stage5: Testing**

- After the code is generated, it is **tested against the requirements to make sure that the products are solving the needs** addressed and gathered during the requirements stage.
- During this stage, **unit testing, integration testing, system testing, acceptance testing** are done.



➤ **Stage6: Deployment**

- Once the software is certified, and no bugs or errors are stated, then it is deployed.
- Then based on the assessment, the **software may be released** as it is or with suggested enhancement in the object segment.
- After the software is deployed, then its maintenance begins.

➤ **Stage7: Maintenance**

- Once when the client starts using the developed systems, then the real issues come up and requirements to be solved from time to time.
- This procedure where the care is taken for the developed product is known as maintenance.



Software Development Process Model

- **Software development models** are **various processes or methods** that are chosen for project development depending on the **objectives and goals of the project**.
- Many development life cycle models have been developed to **achieve various essential objectives**.
- Models **specify the various steps of the process and the order in which they are executed**.



1. Waterfall Model

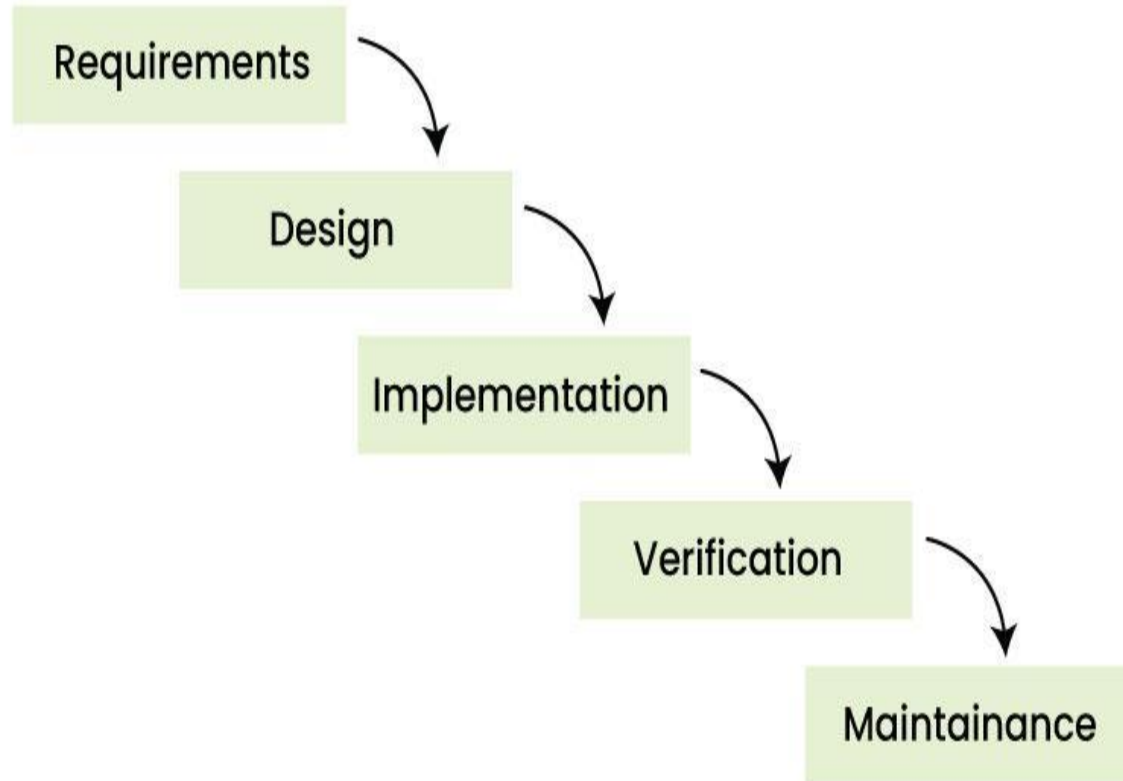
- Waterfall model is a famous and good version of SDLC(System Development Life Cycle) for software engineering.
- The waterfall model is a **linear and sequential model**, which means that a **development phase cannot begin until the previous phase is completed.**
- **We cannot overlap phases** in waterfall model.
- We can imagine waterfall in the following way
“**Once the water starts flowing over the edge of the cliff, it starts falling down the mountain and the water cannot go back up.**”



- Similarly waterfall model also works, once one phase of development is completed then we move to the next phase but cannot go back to the previous phase.
- In the waterfall model, the **output of one phase serves as the input for the other phase.**



○ Phases of Waterfall model



1. Requirement phase:-

- Requirement phase is the first phase of the waterfall model.
- In this phase the requirements of the system are collected and documented.
- This phase is very crucial because the next phases are based on this phase.

2. Design phase:-

- Design phase is based on the fact how the software will be built.
 - The main objective of the design phase is to prepare the blueprint of the software system so that no problems are faced in the coming phases and solutions to all the requirements in the requirement phase are found.
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3. Implementation phase:-

- In this phase, hardware, software and application programs are installed and the database design is implemented.
- Before the database design can be implemented, the software has to go through a testing, coding, and debugging process.
- This is the longest lasting phase in waterfall.

4. Verification phase:-

- In this phase the software is verified and it is evaluated that we have created the right product.
- In this phase, various types of testing are done and every area of the software is checked.
- It is believed that if we do not verify the software properly and there is any defect in it then no one will use it, hence verification is very important.
- One advantage of verification is that it reduces the risk of software failure.



5. Maintenance phase:-

- This is the last phase of waterfall.
- When the system is ready and **users start using** it, then the **problems that arise** have to be **solved time-to-time**.
- **Taking care of the finished software and maintaining it as per time is called maintenance.**



➤ Advantages of Waterfall Model:

- This model is simple and easy to understand.
- This is very useful for small projects.
- This model is easy to manage.
- Each phase of this model is well explained.
- It provides a structured way to do things.

➤ Disadvantages of Waterfall Model

- In this model, complete and accurate requirements are expected at the beginning of the development process.
- We cannot go back to the previous phase due to which it is very difficult to change the requirements.
- In this the testing period comes very late.
- Due to its sequential nature this model is not realistic in today's world.
- This is not a good model for large and complex projects.

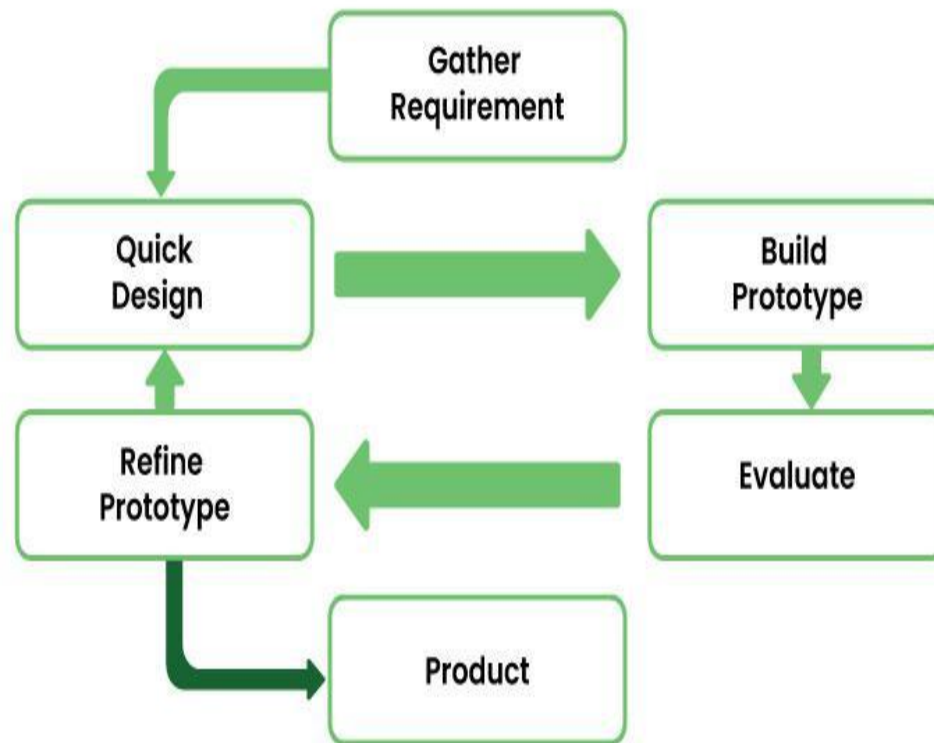


2. Prototype Model

- In software engineering, a prototype is an early version of a software product that is used to test concepts, gather feedback, and refine the design before the final product is released
- Prototype model is an activity in which **prototypes of software applications are created.**
- **First a prototype is created** and then the final product is manufactured based on that prototype.
- This model is **created when we do not know the requirements well.**



- The specialty of this model is that this model **can be used with other models as well as alone.**
- One problem in this model is **that if the end users are not satisfied with the prototype model, then a new prototype model is created again, due to which this model consumes a lot of money and time.**



➤ **The prototype model has the following phases**

1. **Requirement gathering:** The first step of prototype model is to **collect the requirements**, although the **customer does not know much about the requirements** but the major requirements are defined in detail.
2. **Build the initial prototype:** In this phase the **initial prototype is built**. In this some basic requirements are displayed and user interface is made available.
3. **Review the prototype:** When the **construction of the prototype is completed**, it is **presented to the end users or customer** and **feedback is taken from them about this prototype**.
 - This feedback is used to **further improve the system and possible changes** are made to the prototype.
4. **Revise and improve the prototype:** When feedback is taken from end users and customers, the **prototype is improved on the basis of feedback**.
 - If the customer is not satisfied with the prototype, **a new prototype is created and this process** continues until the customer gets the prototype as per his desire.

➤ Advantages of Prototype model

- Prototype Model is suggested to create **applications whose prototype is very easy** and which always includes human machine interaction within it.
- When we know only the **general objective of creating software, but we do not know anything in detail about input, processing and output**. Then in such a situation we make it a Prototype Model.
- When a **software developer is not very sure about the capability of an algorithm or its adaptability to an operating system**, then in this situation, using a prototype model can be a better option.



➤ Disadvantages of Prototype model

- The customer often requests **small fixes and changes in the first version of the prototype** model rather than rebuilding the system.
- The first version of the Prototype Model **often involves compromises.**
- Developers may **make compromises in implementation to quickly get the prototype running.**

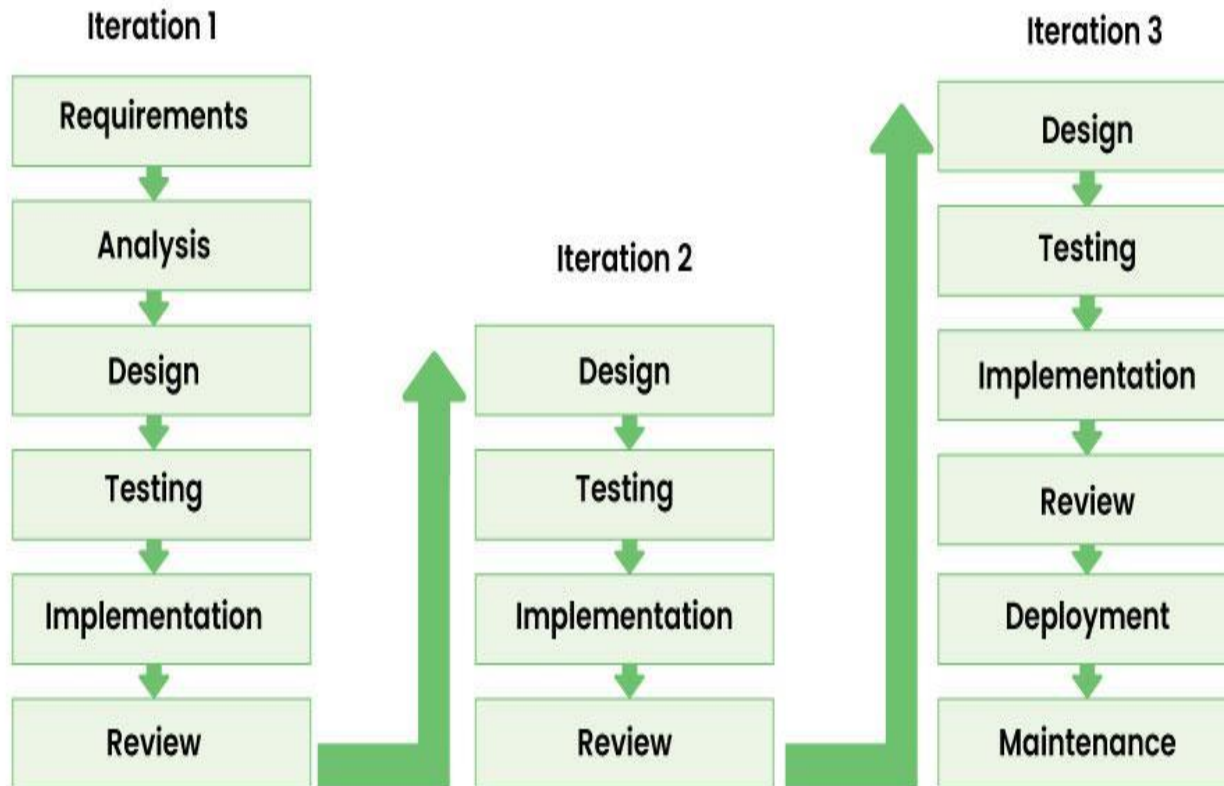


3. Iterative Model

- In Iterative model we start developing the software with **some requirements and when it is developed, it is reviewed.**
 - If there are **requirements for changes in it, then we develop a new version** of the software based on those requirements.
 - This **process repeats itself many times until we get our final product.**
 - So, in Iterative model a software is developed by following several iterations. **Iteration means that we are repeating the development process again and again.**
 - For example, we develop the first version of the software following the SDLC process with some software requirements.
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- After the first version is developed, if there is a need to change the software, then a new version is developed with the second iteration. Now again we will see if the new version is enough, if not then we will make changes in it with the third iteration. The iteration will be repeated until the complete software is ready.
- The basic concept of Iterative model is that the software should be developed through repeated cycles or what we also call iteration and only a small part of it should be developed at a time.
- This model was developed to overcome the drawbacks of the classical waterfall model.





➤ Phases of iterative model

1. **Requirement gathering & analysis:** In this phase, all the software requirements of the customer are collected and it is analyzed whether those requirements can be meet or not. Besides, it is also checked whether this project will not go beyond our budget.
2. **Design:** In this phase the design of software is prepared. For this, various diagrams like Data Flow diagram, class diagram, activity diagram, state transition diagram, etc. are used.
3. **Implementation:** Now the design of software is implemented in coding through various programming languages. We also call this coding phase.



4. **Testing:** After the coding of the software is done, it is now tested so that the bugs and errors present in it can be identified. To do this, various testing techniques like performance testing, security testing, requirement testing, stress testing, etc. are done

.

5. **Deployment:** Finally the software is given to the customer. After this the customer starts using that software in his work environment.

6. **Review:** After the software is deployed in its work environment, it is reviewed. If any error/bug is found or any new requirements come in front of developer, then again these phases are repeated with new iteration and a new version is developed.

7. **Maintenance:** In this phase we look at customer feedback, solve problems, fix errors, update software, etc.



➤ Advantage of Iterative model

- In iterative models, bugs and errors can be identified quickly.
- Under this model, software is prepared quickly with some specifications.
- Testing and debugging the software becomes easier during each iteration.
- We get reliable feedback from users along with blueprints.
- This model is easily adaptable to constantly changing needs.
- Risks are identified and resolved during iteration.

➤ Disadvantage of Iterative model

- Iterative model is not suitable for small projects.
- Since we have to repeat iterations many times in the software development process due to which we require more resources.
- Since the requirements are constantly changing, we have to make frequent changes in the software.
- Due to constantly changing requirements, the budget of the project also increases and it takes more time to complete it.
- It is very difficult to tell by what date the complete software will be ready.



1. Spiral Model

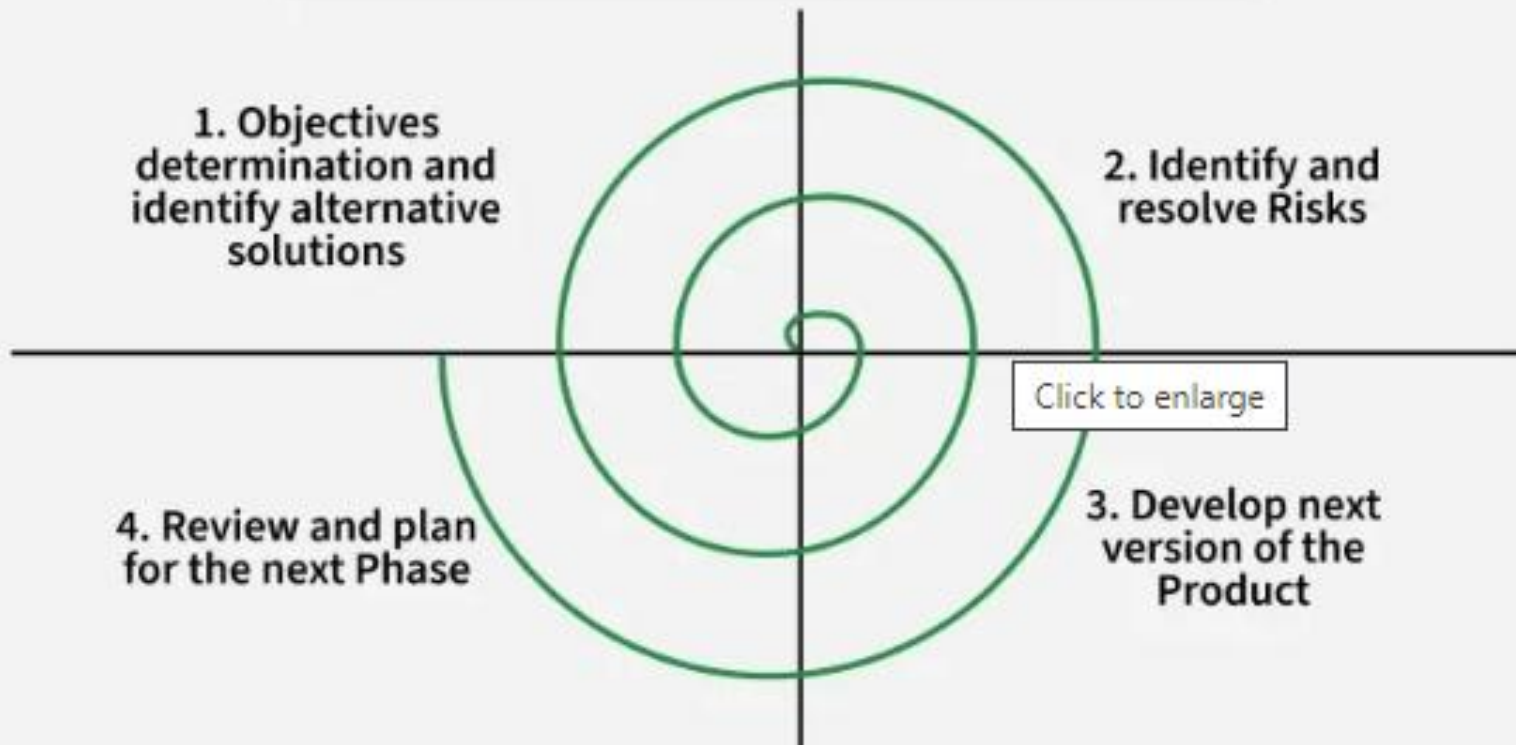
- Spiral model is a software development process model. This model has characteristics of both iterative and waterfall models.
- This model is used in projects which are large and complex.
- This model was named spiral because if we look at its figure, it looks like a spiral, in which a long curved line starts from the center point and makes many loops around it.
- The number of loops in the spiral is not decided in advance but it depends on the size of the project and the changing requirements of the user.
- We also call each loop of the spiral a phase of the software development process.



- A software project goes through these loops again and again in iterations.
- After each iteration a more and more complete version of the software is developed.
- The most special thing about this model is that risks are identified in each phase and they are resolved through prototyping. This feature is also called Risk Handling.
- Since it also includes the approaches of other SDLC models, it is also called Meta Model.
- It was first developed by Barry Boehm in 1986.



Phases of the Spiral Model



➤ Those phases are as follows:-

- **Determining objectives and alternate solutions:** In the first phase, whatever requirements the customer has related to the software are collected. On the basis of which objectives are identified and analyzed and various alternative solutions are proposed.
- **Identifying and resolving risks:** In this phase, all the proposed solutions are assessed and the best solution is selected. Now that solution is analyzed and the risks related to it are identified. Now the identified risks are resolved through some best strategy.
- **Develop and test:** Now the development of software is started. In this phase various features are implemented, that is, their coding is done. Then those features are verified through testing.
- **Review and plan for the next phase:** In this phase the developed version of the software is given to the customer and he evaluates it. Gives his feedback and tells new requirements. Finally planning for the next phase (next spiral) is started.

➤ Advantages of Spiral Model

- Spiral model is suitable for large and complex projects.
- It is easy to estimate how much the project will cost.
- Risk analysis is done in each phase of this model.
- The customer can see the look of his software only in the early stages of the development process.
- Since continuous feedback is taken from the customer during the development process, the chances of customer satisfaction increases.

➤ Disadvantage of Spiral Model

- This is the most complex model of SDLC, due to which it is quite difficult to manage.
 - This model is not suitable for small projects.
 - The cost of this model is quite high.
 - It requires more documentation than other models.
 - Experienced experts are required to evaluate and review the project from time to time.
- 